

# Compatibility policy

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This policy defines how long BackupSage's public contracts remain compatible and how a future breaking change is announced. It applies to released v1.x versions; unreleased main-branch behavior is not a compatibility promise.

## Window and change rules

Within v1.x, every surface listed below is stable. Additions are permitted where stated; removals, renames, semantic changes, and incompatible format changes are not. A v1.x release must continue to read the on-disk data formats created by earlier v1.x releases.

A breaking change requires a new major version. Before that major release, BackupSage will document the affected surface, migration or re-indexing steps, and the first release containing the change in both this policy and its release notes. A deprecation notice may be published during v1.x, but it does not make the v1.x contract breakable.

## Public surfaces

| Surface          | Current form                                                                                                                                                                                                                      | v1.x promise                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| JSON reports     | <code>dedup --json report</code><br><code>"version": 1;</code><br><code>search --json,</code><br><code>search --all --</code><br><code>json, master list --</code><br><code>json, and master</code><br><code>verify --json</code> | Stable and additive-only. Existing keys retain their names, types, and meanings; <code>dedup</code> remains report version 1. v1.0.2 added the keeper-star fields (#9): <code>actionable</code> , <code>review_only_bytes</code> , <code>transitive_only_files</code> , <code>actionable_rule</code> — additive. In the same release <code>reclaimable_bytes / duplicate_files</code> stopped counting transitive-only members (a correctness fix to match their documented meaning: only members measured safe against the keeper are reclaimable). |
| Exit codes       | <code>0</code> clean, <code>1</code> error, <code>2</code> completed with skips; clap usage errors currently also use <code>2</code>                                                                                              | Stable meanings, including the documented usage-error caveat.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Per-source index | SQLite schema version <code>3</code>                                                                                                                                                                                              | Stable within v1.x. v1.x continues to read schema-v3 indexes and the supported schema-v2 indexes, which remain v2-limited where metadata required by a command is absent.                                                                                                                                                                                                                                                                                                                                                                            |
| Master catalog   | SQLite master catalog containing registered index metadata and replicas                                                                                                                                                           | Stable within v1.x. A catalog created by an earlier v1.x release remains readable by later v1.x releases; registration and query behavior preserve existing entries.                                                                                                                                                                                                                                                                                                                                                                                 |
| Perceptual hash  | <code>sage-dct-v1</code> 64-bit DCT pHash                                                                                                                                                                                         | Frozen. The algorithm identifier and its hash semantics do not change; any future algorithm receives a new identifier, and mixed algorithms are not compared.                                                                                                                                                                                                                                                                                                                                                                                        |
| CLI              | Documented commands, subcommands, flags, defaults, and accepted values                                                                                                                                                            | Stable within v1.x. New commands or flags may be added, but existing invocation syntax and behavior are not removed or repurposed. <code>dedup --threshold</code> remains capped at <code>0.3</code> , because its four 16-bit bands guarantee recall only through Hamming distance 3.                                                                                                                                                                                                                                                               |

`docs/CONTRACT.md` records the exact current JSON and exit-code shapes and the golden tests that enforce them. This policy adds the version window and deprecation procedure; it does not widen a surface that is documented there as human-oriented and unstable.

## On-disk data

BackupSage never rewrites an existing index or master catalog merely to read it. A later v1.x release may create a newer index only through the normal explicit indexing flow. If a future major version cannot read an old format, it will provide the migration or re-indexing instructions required by the deprecation procedure above before that breaking release ships.